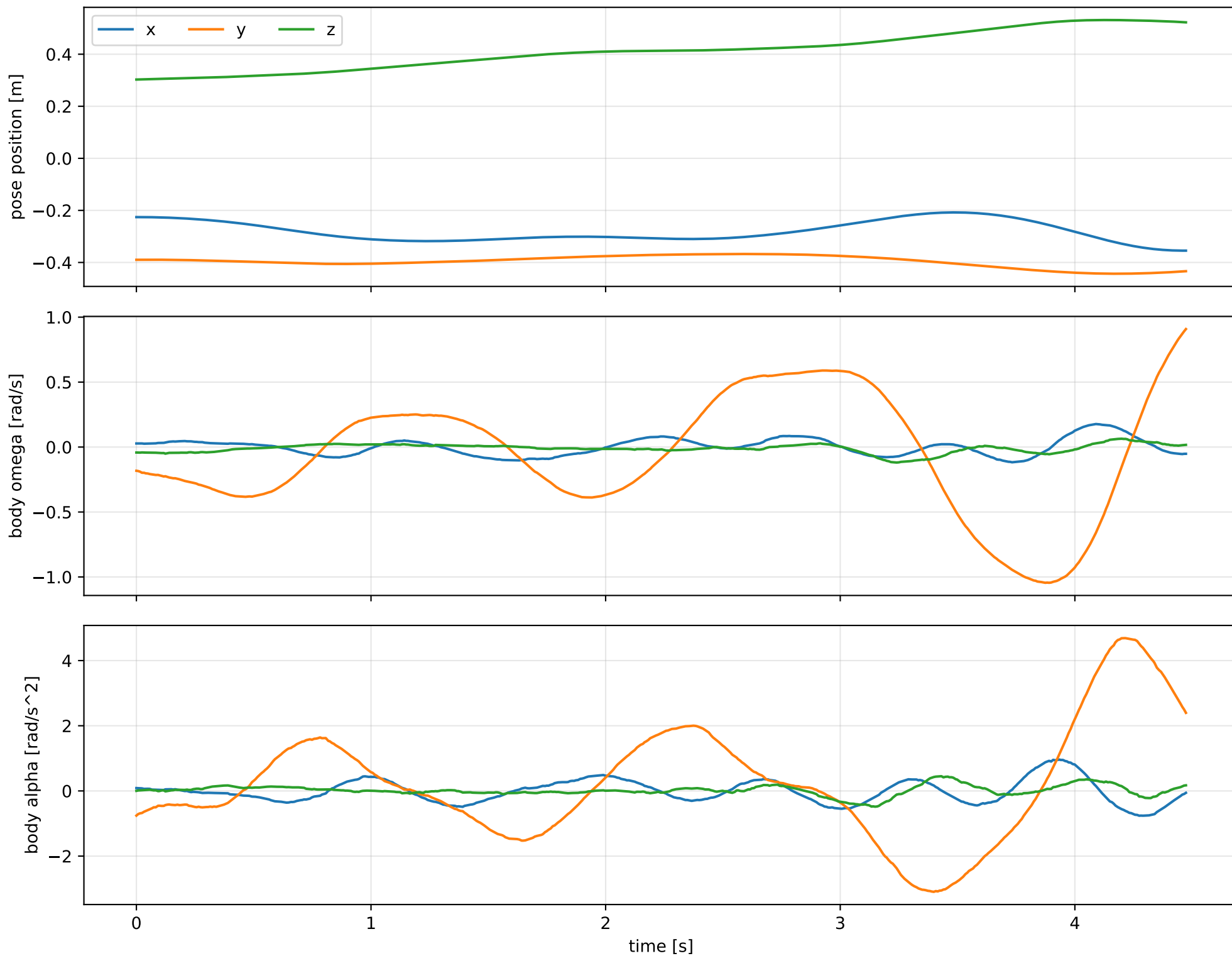
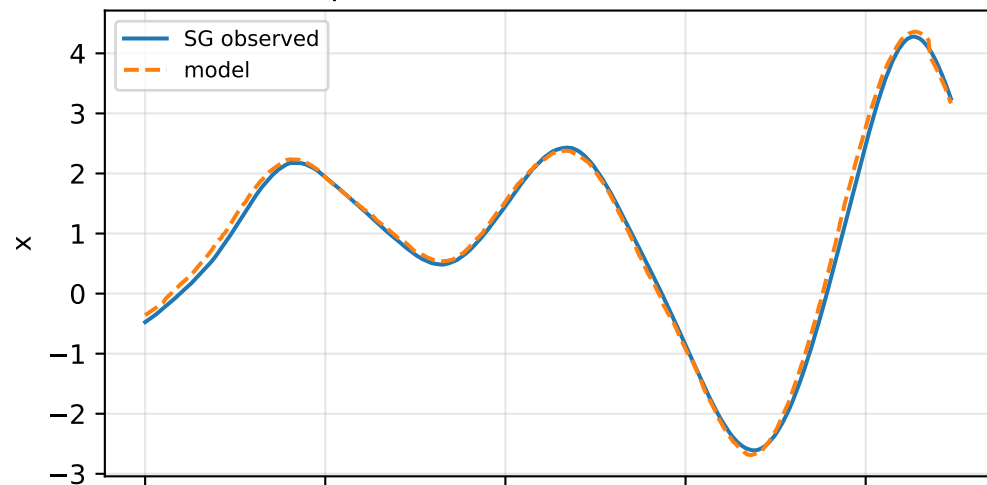
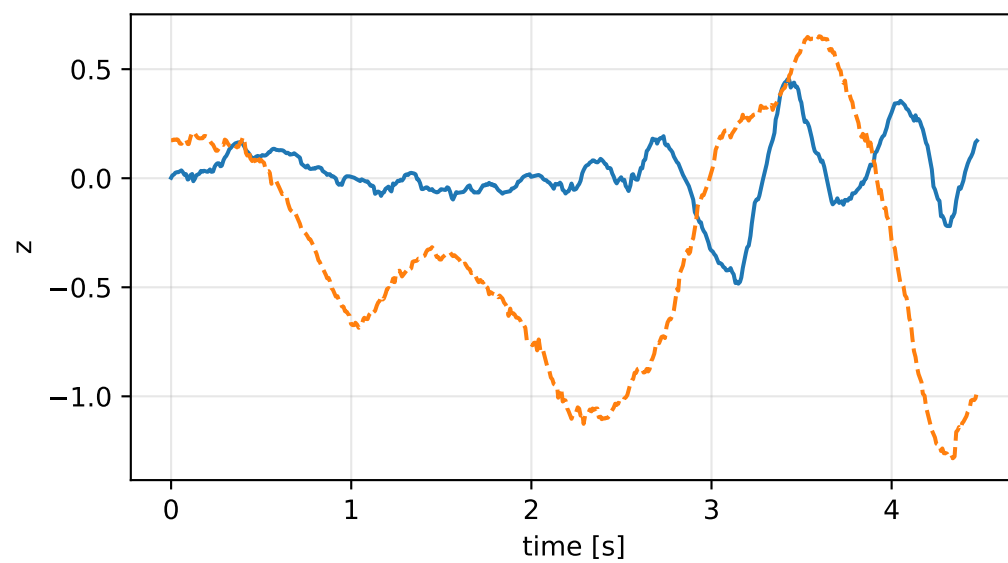
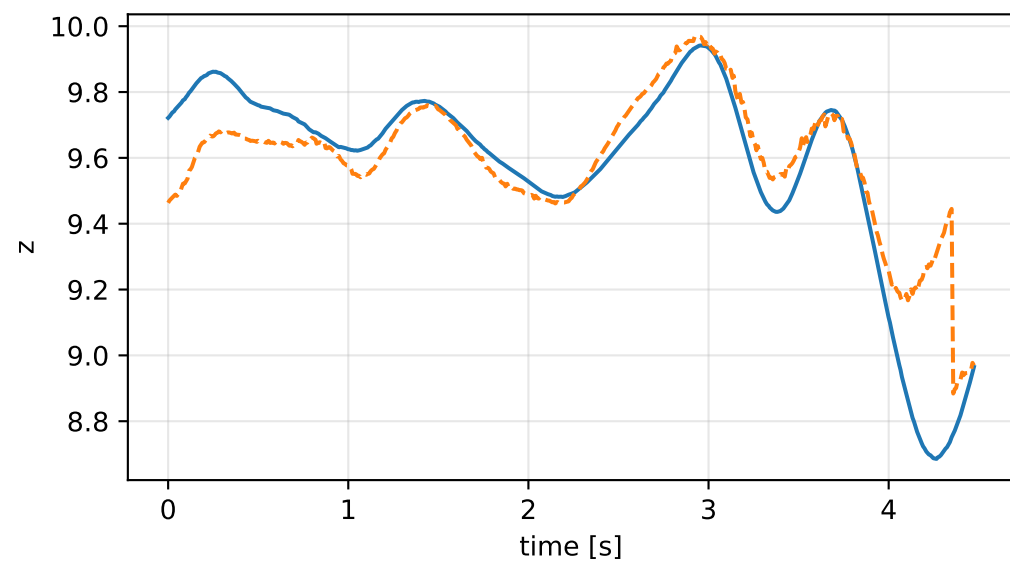
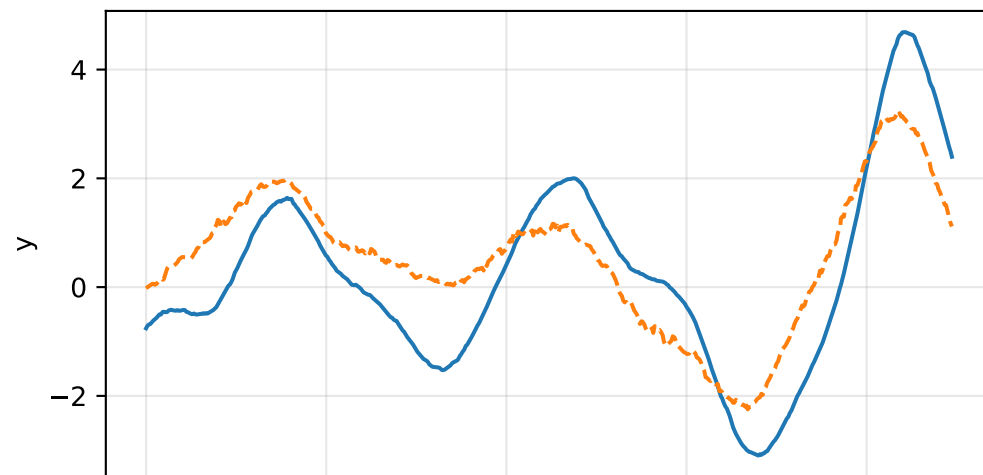
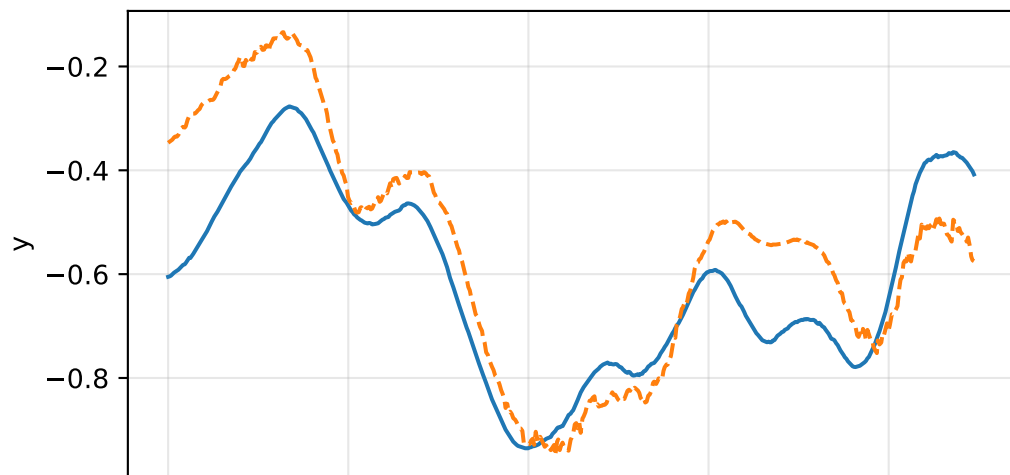
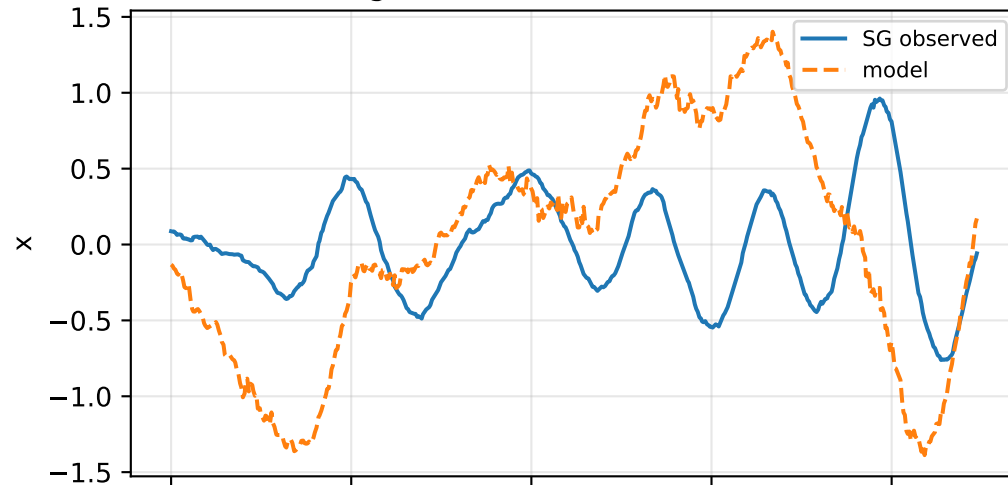


cov_block_s_alpha: SG trajectory and derived motion

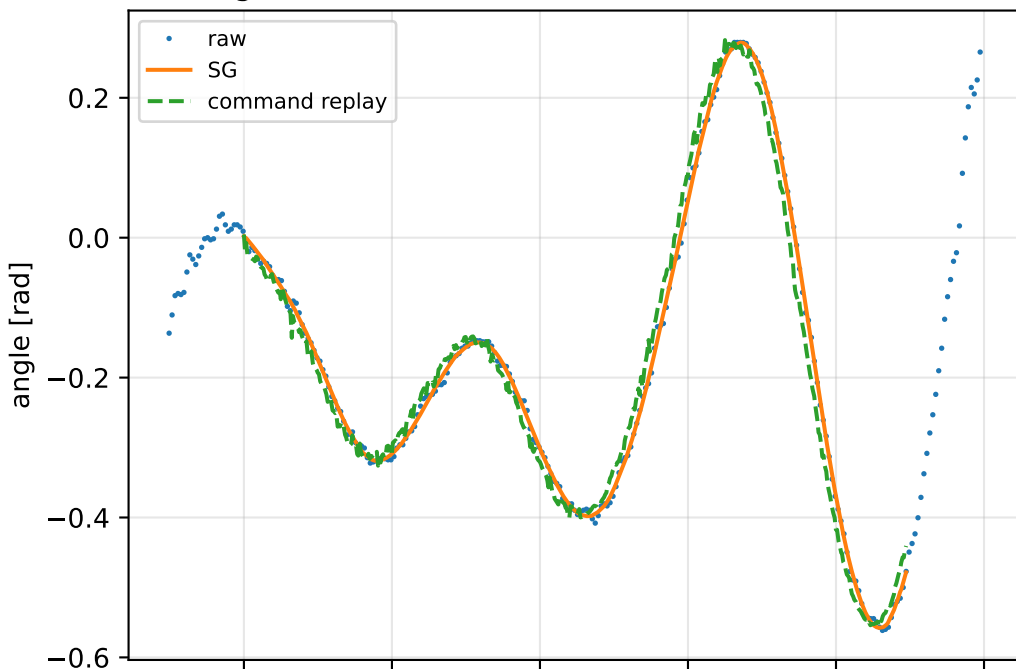


cov_block_s_alpha: acceleration residual objective

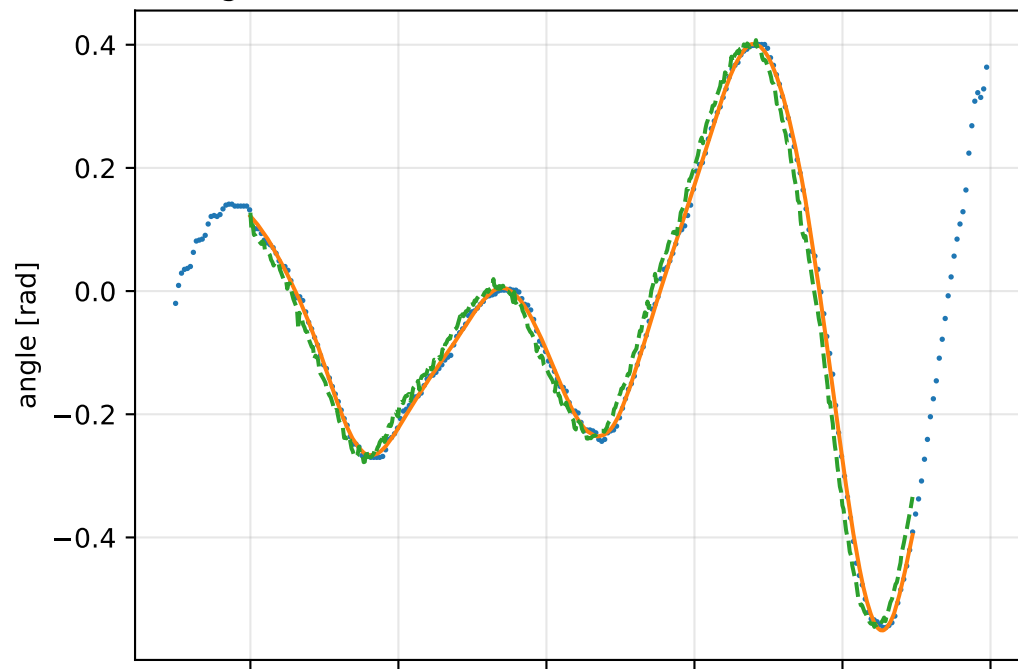
specific acceleration [m/s²]angular acceleration [rad/s²]

cov_block_s_alpha: gimbal measurement audit (objective=measured_sg)

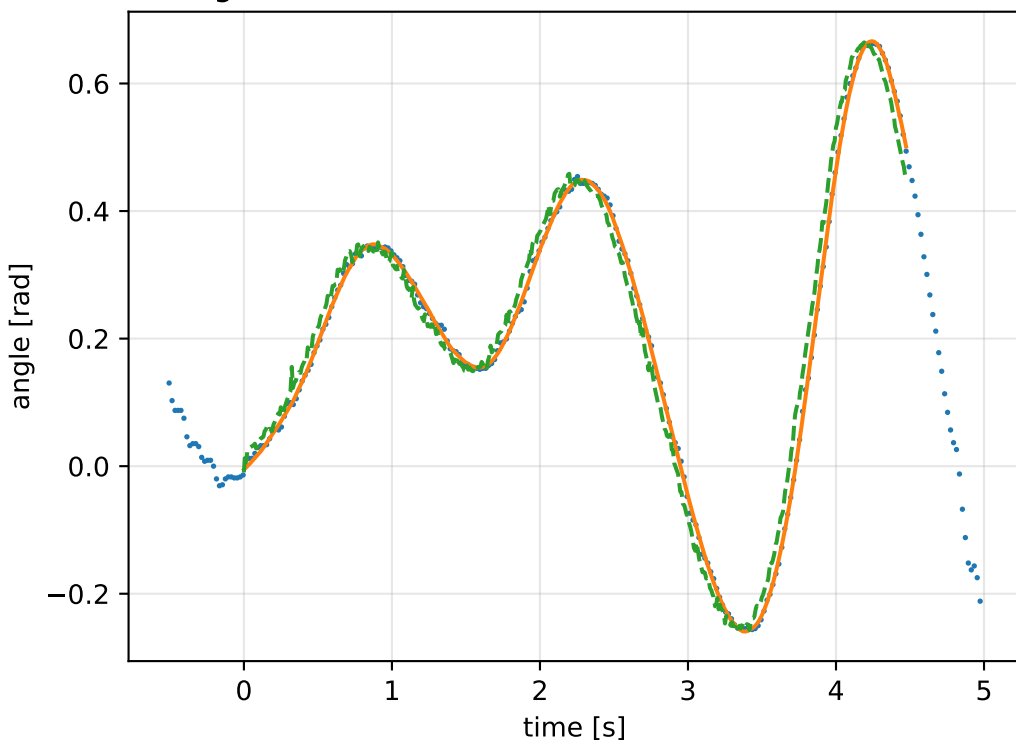
gimbal 1: RMSE=0.027, max=0.07529 rad



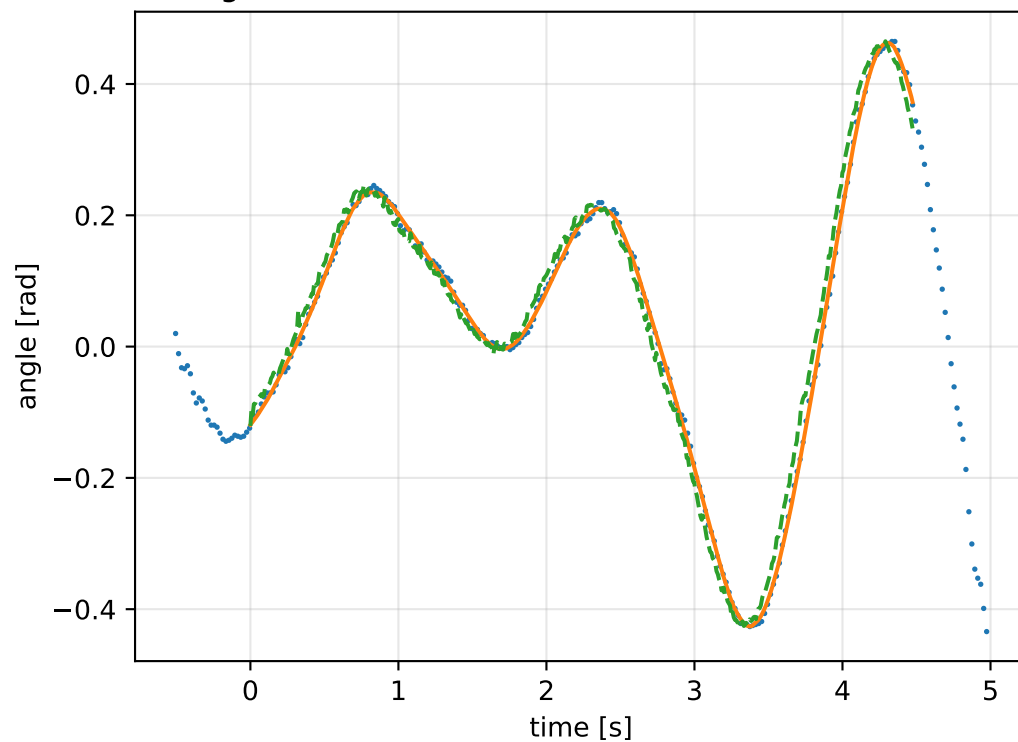
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

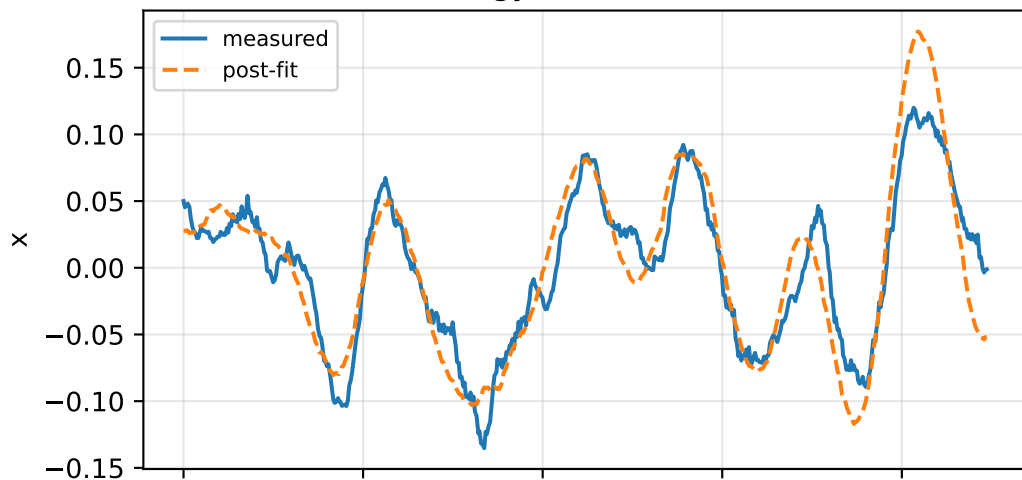


gimbal 4: RMSE=0.02598, max=0.07261 rad

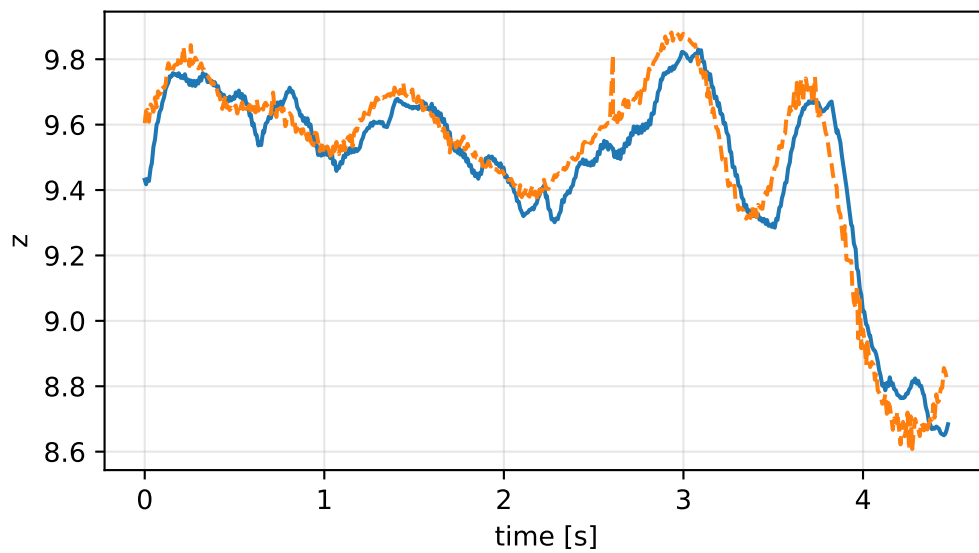
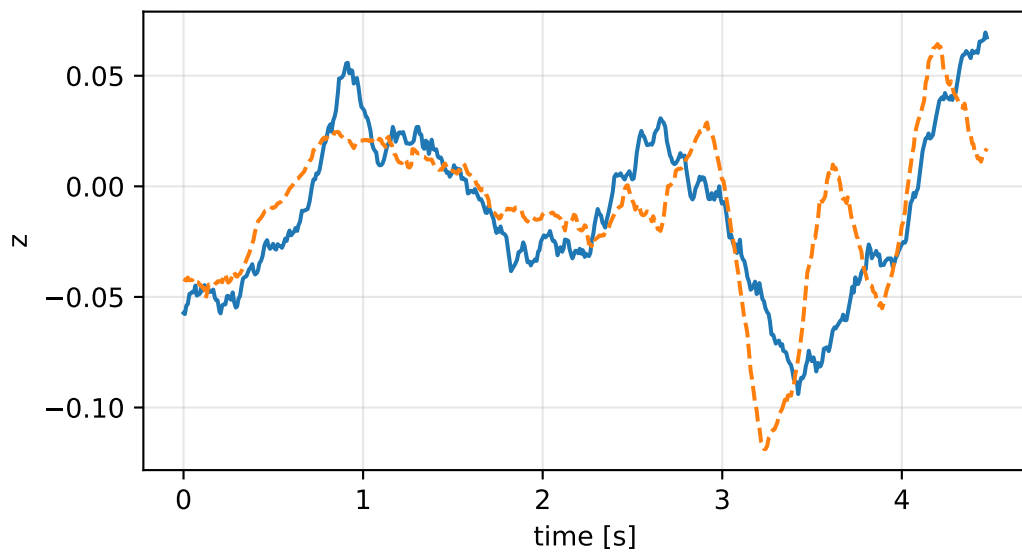
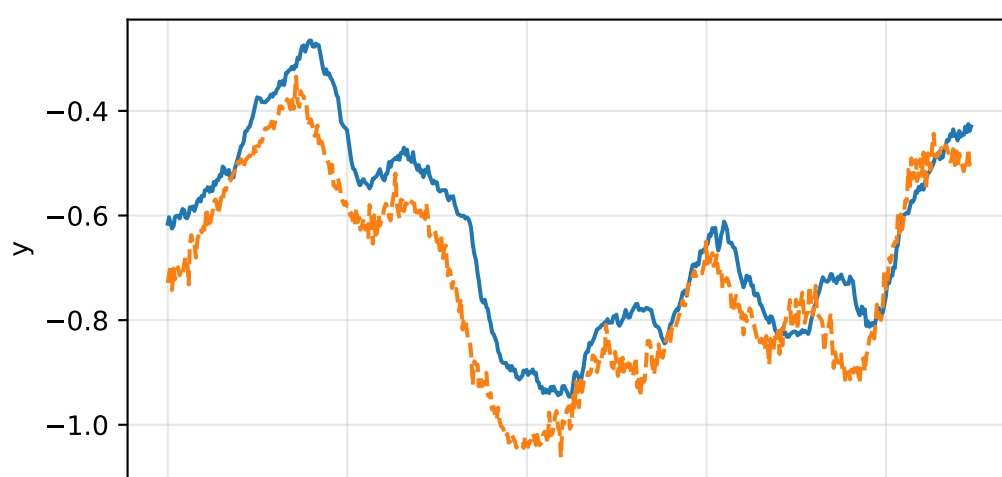
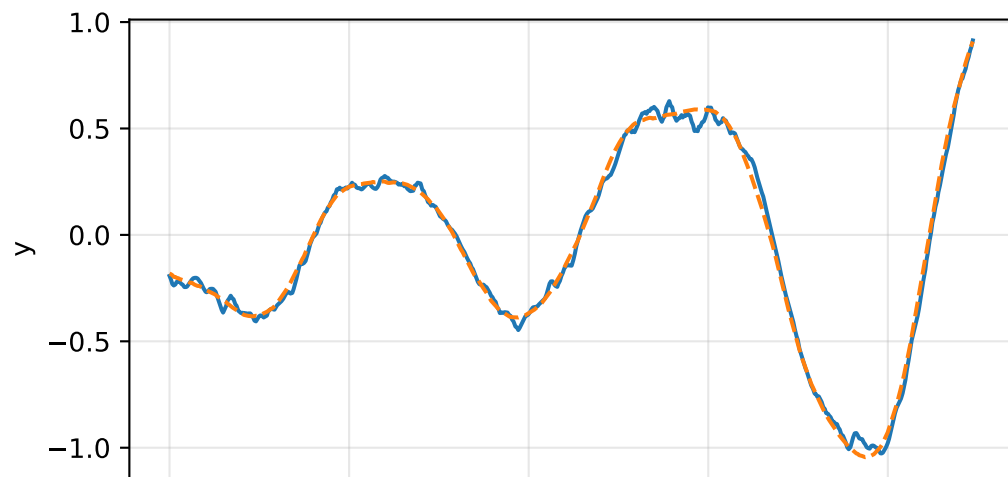
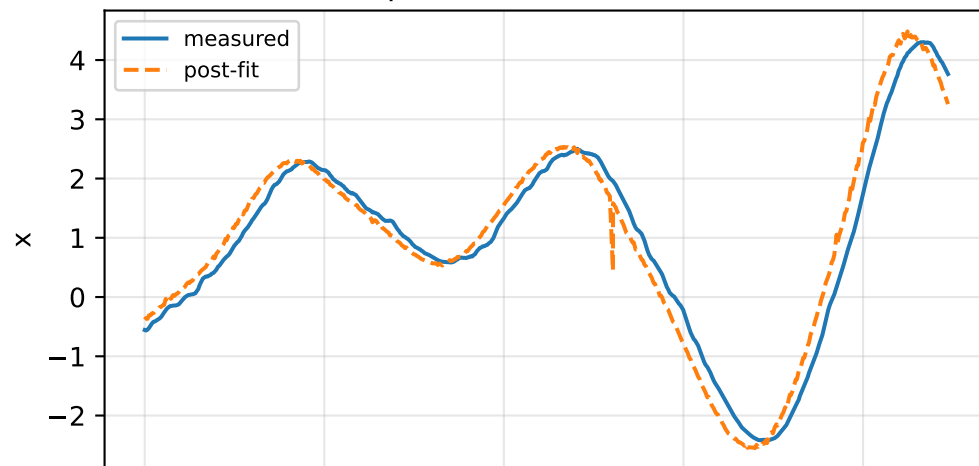


cov_block_s_alpha: IMU comparison (diagnostic only)

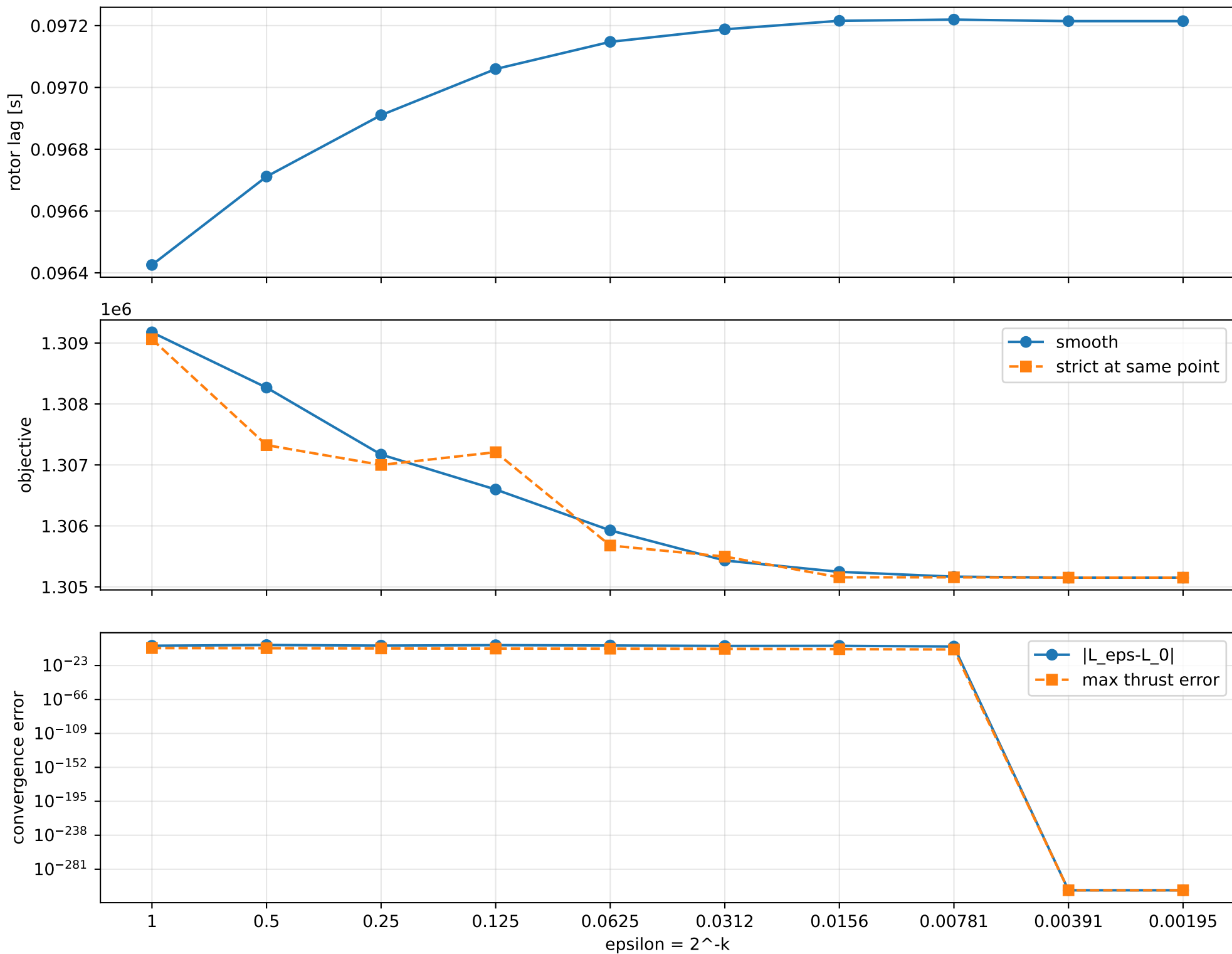
gyro [rad/s]



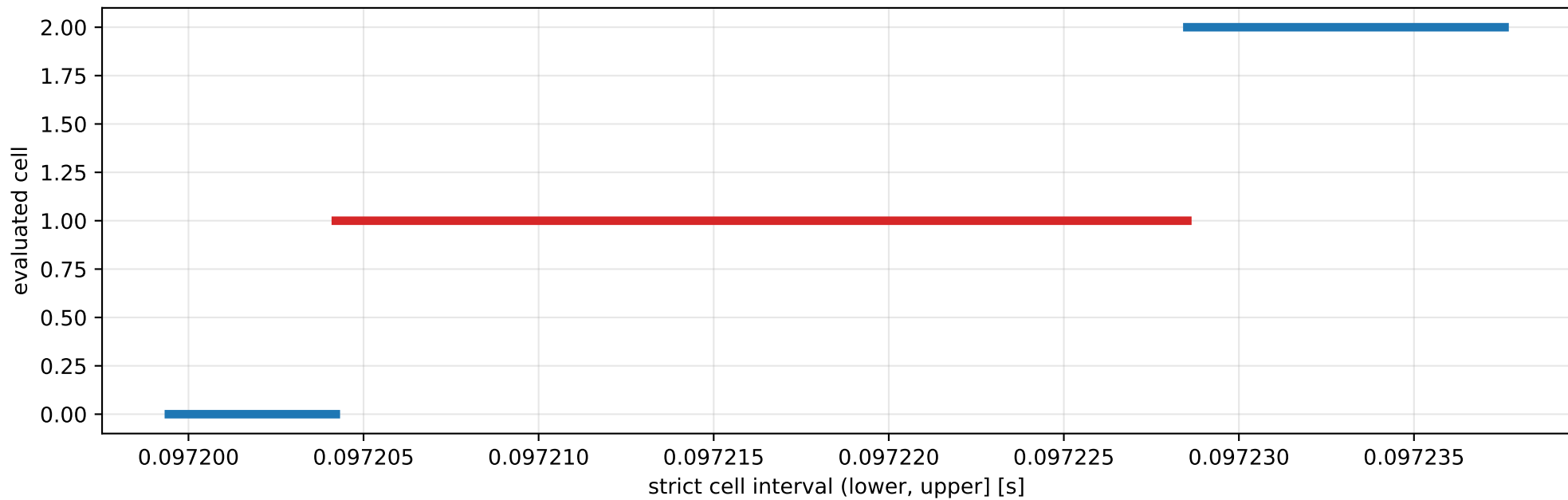
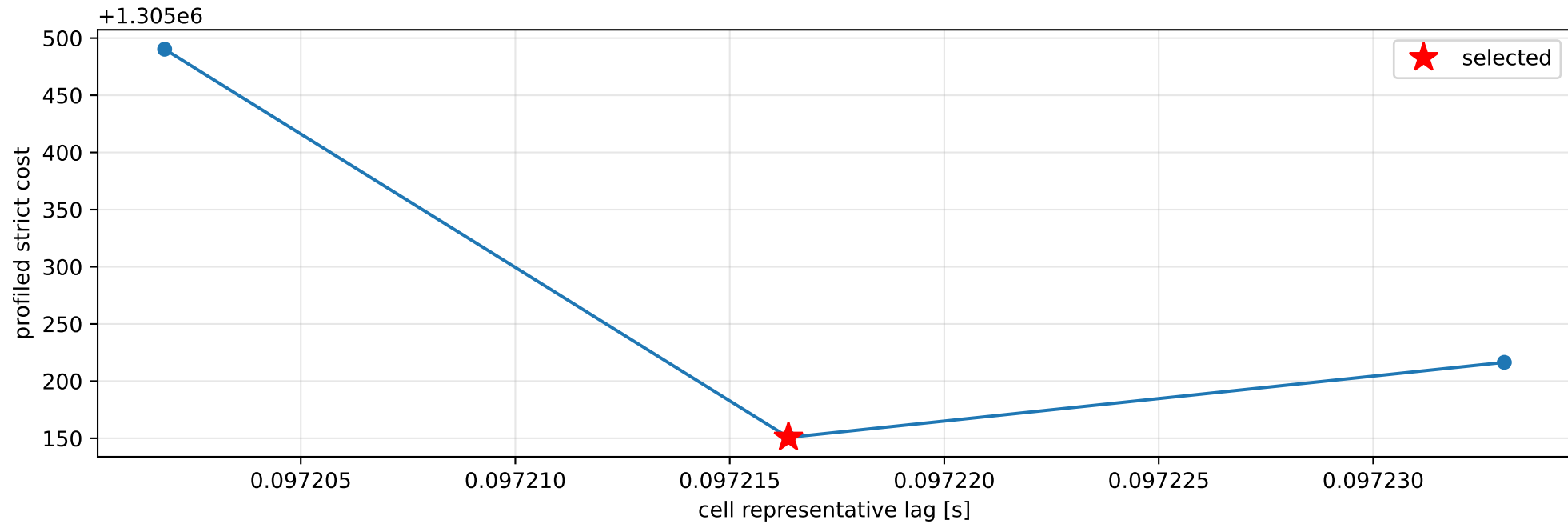
specific force [m/s^2]



cov_block_s_alpha: smooth-to-strict continuation



cov_block_s_alpha: exact strict-ZOH lag cells



selected (0.0972042084, 0.0972285271] s; final smooth 0.0972145849 s; data boundary=False

cov_block_s_alpha: parameter estimate

Case: cov_block_s_alpha

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 16.0515246

inertia [kg m²]:

```
[[ 2.33292018  1.84078468  0.91334565]
 [ 1.84078468  3.24148861 -0.95258177]
 [ 0.91334565 -0.95258177  4.58273261]]
```

CoG body [m]: [0.09302122 0.15448257 -0.09732149]

force effectiveness: [3.20930860e-05 3.92728518e+00 1.27148687e+01 6.70075548e+00]

scale-free J/m [m²]:

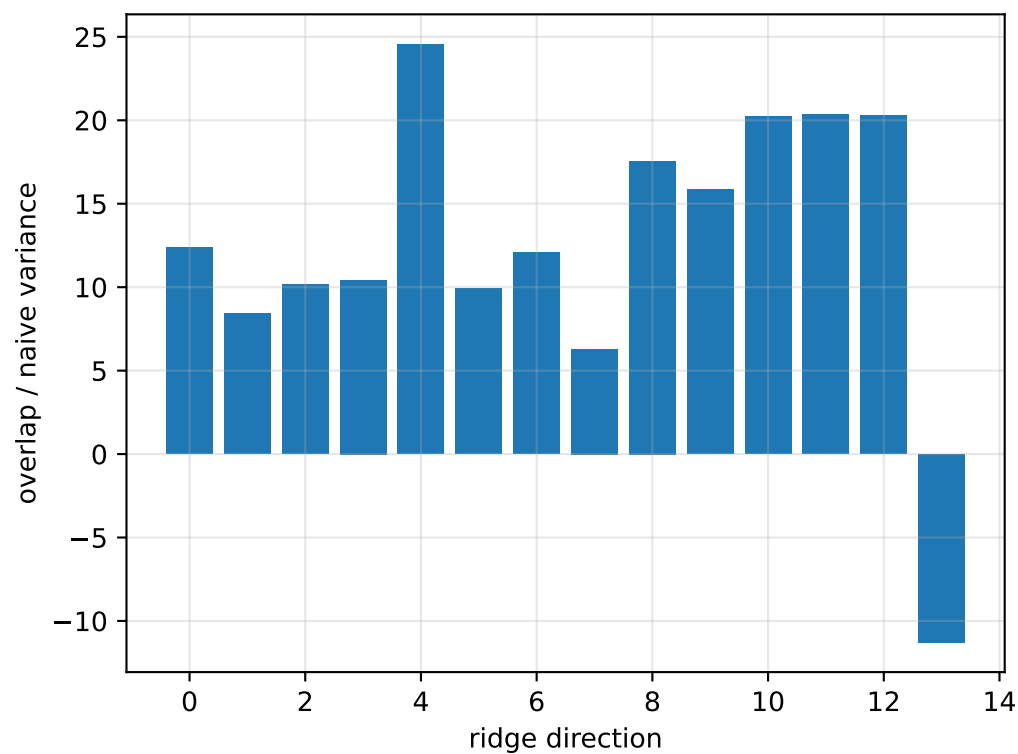
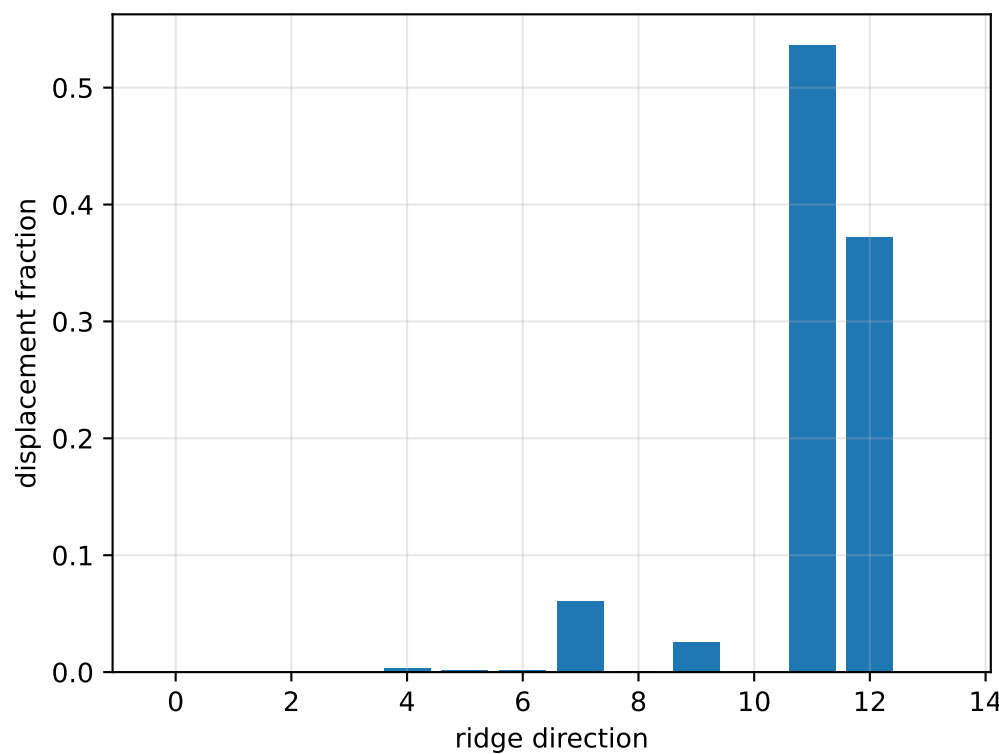
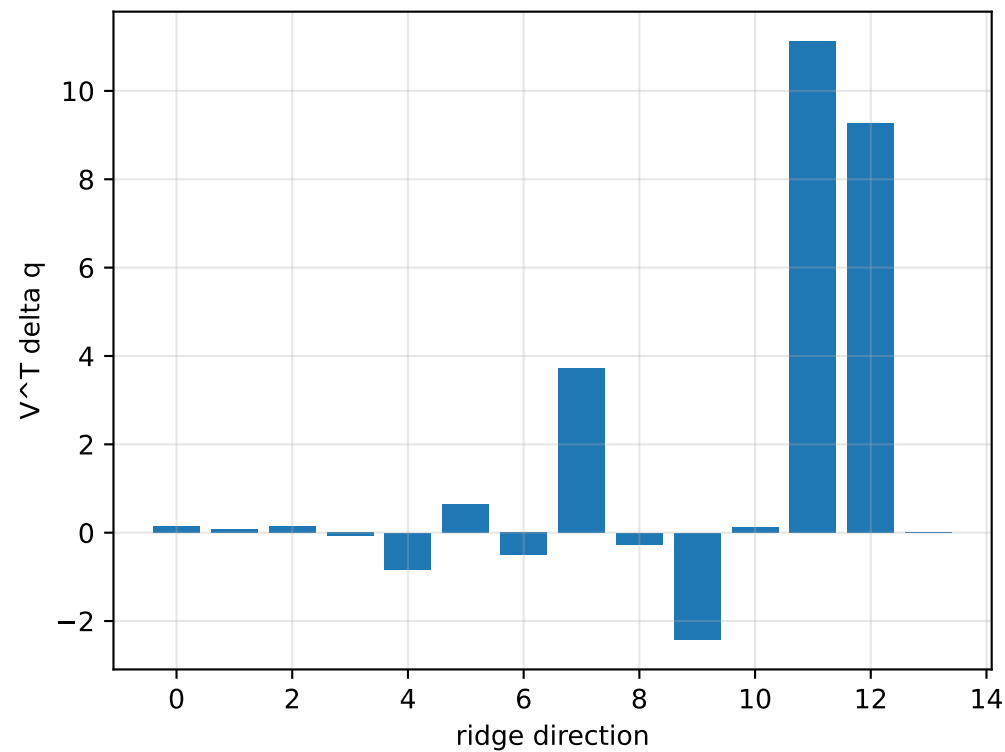
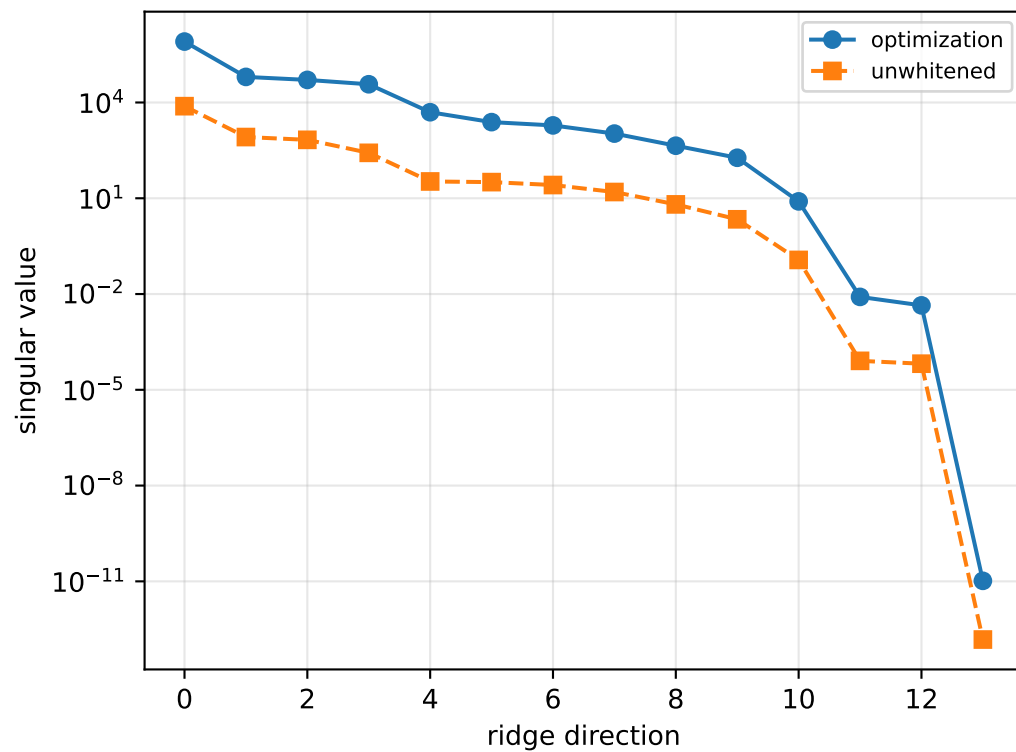
```
[[ 0.14533948  0.11467974  0.05690087]
 [ 0.11467974  0.20194272 -0.05934525]
 [ 0.05690087 -0.05934525  0.28550139]]
```

scale-free f/m: [1.99937930e-06 2.44667424e-01 7.92128409e-01 4.17452898e-01]

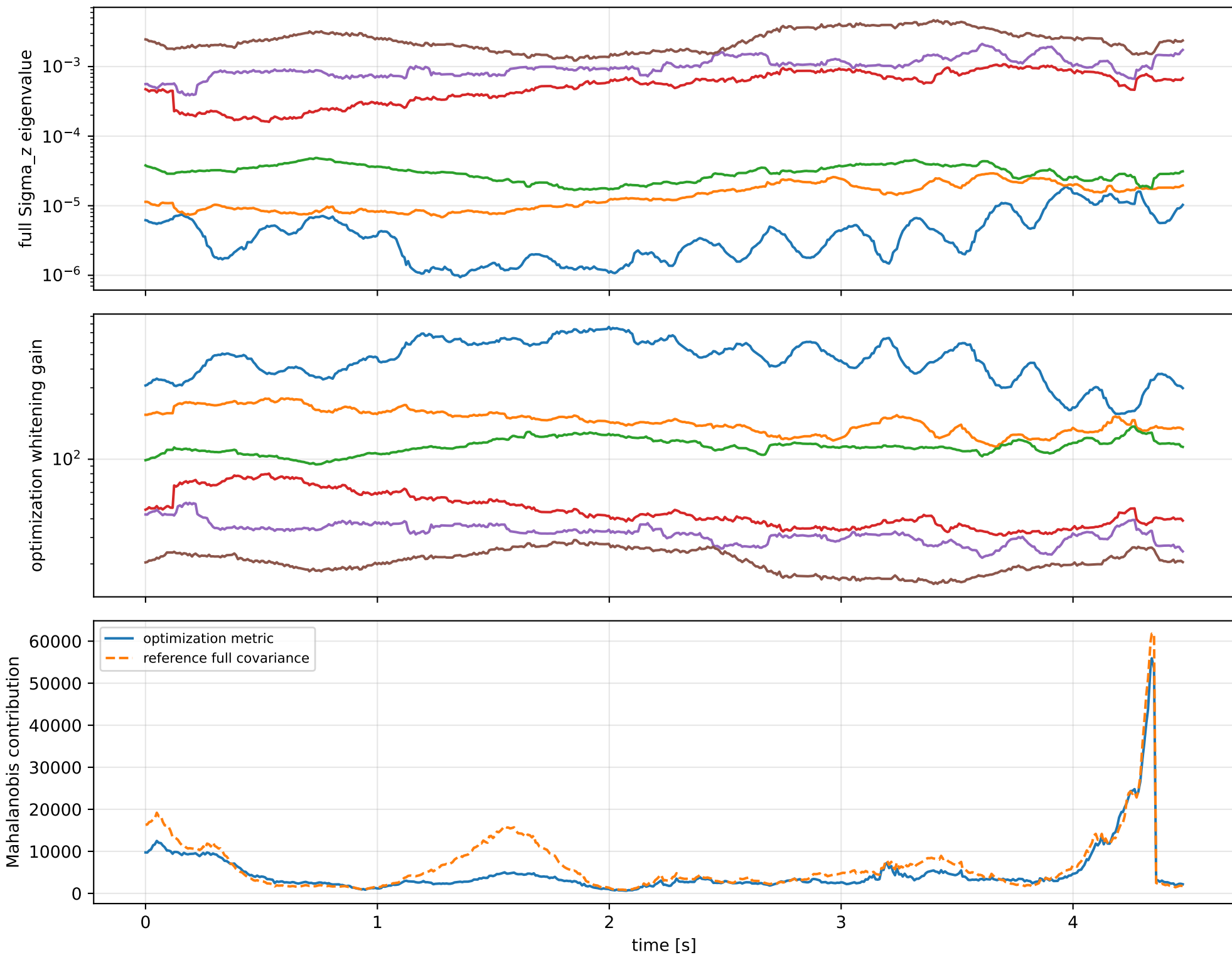
rotor lag cell: (0.0972042084, 0.0972285271] s

representative: 0.0972163677 s

cov_block_s_alpha: ridge spectrum (rank 13, nullity 1, ||v||=2.25e-11)



cov_block_s_alpha: optimization vs reference covariance



cov_block_s_alpha: wrench / closure (force max 5.42e-14, torque max 3.11e-15)

